

RL-based Soft Manipulator Controller for Cleaning Tasks in Confined Environments

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1 The (Robotic) Motivation Behind the Specific Problem Considered (WHY?)

Nowadays manipulators are increasingly used to assist or replace human workers in performing tasks under extreme conditions, such as high-altitude, high-radiation and high voltage environment where direct manual operation may involve substantial risks. However, cleaning tasks are usually required in these scenarios as a part of inspection and routine maintenance. Conventionally, Rigid manipulators are widely deployed in these scenarios. However, their performance can be significantly limited in narrow, confined, or geometry complex spaces due to their low compliance. Moreover, rigid collision between the manipulator and environment may lead to irreversible damage to either the equipment or robot. Soft manipulators offer a promising alternative because their highly compliant body allow them to deform flexibly and maneuver through the narrow places while also reduce the damage caused by the rigid contact. In this project, we mostly focused on developing a octopus-arm-like soft manipulator capable of traversing a narrow opening in a wall and subsequently perform the clean task on the opposite side.

2 Current Conventional Algorithms/Methods and Their Limitations

Conventional methods for motion-control include methods such as PID control and model predictive control (MPC), both of which rely on an appropriate model of the controlled system to varying degrees. However, the continuum structure and deformable body make it difficult for soft manipulator to have an accurate analytical model to describe the dynamic system. As a result, the conventional methods can be challenging. Some approximate methods such as piecewise constant curvature (PCC) and finite element models (FEM) can be used to describe the dynamic of soft manipulators. But there exist a trade off between the computational efficiency and modeling accuracy for real-time control. Therefore, We propose using deep reinforcement learning to allow the manipulator to learn an effective control policy by interacting with the simulation environment. Besides, the neural-network-based policy can provide a fast online inference. In addition, to facilitate the following sim-to-real work, the observable states provided to the learning-based controller will be limited within the region where the data can be practically measured on a physical robot.

3 Problem Statement (WHAT?)

To train the policy in a simulator, we will build a pseudo-rigid-body model of the soft manipulator in MuJoCo and construct a structured environment around the manipulator. The manipulator will be required to move its end-effector into a confined space, such as a blind box, and perform a cleaning task while avoiding collisions with surrounding obstacles.

We assume that the environment of the work space of our manipulator is fully known. In addition, several IMUs placed along the intermediate sections of the manipulator will be used to estimate its overall shape and configuration. To define the task goal, several markers will be placed inside the confined space. Once the end-effector reaches a target marker with the desired position and orientation, the corresponding target will be considered successfully reached. The task will be considered complete when all required targets have been reached without collision.

4 Reinforcement Learning Formulation (HOW?)

4.1 State Space

The state space includes the estimated shape and configuration of the manipulator derived from IMU measurements. For example, 4–6 IMUs can be distributed along the manipulator to provide orientation information for different sections. In addition, the state representation includes the states of the three cable actuators, the positions of surrounding obstacles, and the relative position and orientation of the next target marker with respect to the end-effector.

4.2 Action Space

Since the manipulator is actuated by three cables, the action is defined as

$$a_t = [\Delta q_1, \Delta q_2, \Delta q_3]$$

where each component represents the displacement command applied to one cable actuator.

4.3 Reward Structure

A small negative reward (e.g., -0.1) is applied at each time step to encourage efficient task completion. A larger penalty (e.g., -5) is imposed when the manipulator collides with surrounding obstacles. A positive reward (e.g., $+10$) is given when the end-effector reaches a target marker within predefined position and orientation tolerances, such as 2cm in position and 10° in orientation. An additional terminal reward (e.g., $+30$) is provided when all required target markers are successfully reached without collision.

4.4 Neuro Network Structure

Since the soft manipulator is continuous and deformable, the states of neighboring segments are spatially coupled rather than independent. To capture these local spatial correlations, we will employ a CNN to extract features from the states of adjacent segments.

In addition, the state of a continuum manipulator is inherently time-dependent, and its current behavior may be influenced by previous configurations and control actions. Therefore, a GRU will be incorporated into the policy network to capture temporal dependencies and construct a history-aware state representation for reinforcement learning.

If time permits, we will also implement an LSTM-based architecture and compare its performance with the GRU-based policy.

5 RL Algorithms to Be Investigated and Expected Outcomes

Since the pseudo-rigid-body model contains hundreds of virtual links and joints, the simulator can provide detailed state information that is not fully observable on the physical robot. To address this issue, we adopt an asymmetric Actor-Critic structure trained using PPO, where the actor only uses realistically measurable observations, while the critic can access the full privileged simulator state during training.

We will investigate whether this asymmetric training strategy improves policy learning compared with a conventional symmetric Actor-Critic structure. We will also evaluate whether the GRU-based temporal model can improve control performance under partial observation from distributed IMUs. The learned policy will be evaluated in terms of task success rate, collision rate, control accuracy, and robustness in confined-space tasks.